Single-exposure Polarization Digital Holographic Microscopy (SPDHM) for Quantitative Stress Metrology in Laser-modified Glass Substrates for Advanced Packaging

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Abstract

Single-exposure polarization digital holographic microscopy (SPDHM) was developed as a non-destructive method for quantitative residual-stress metrology in laser-modified glass substrates for advanced semiconductor packaging. A polarization-resolved off-axis holographic configuration records polarization information in a single exposure, enabling reconstruction of the complex optical field. Angular spectrum propagation provides full-field amplitude and phase at the target plane, while Jones-vector retrieval yields stress-induced birefringence. Circular polarization of both object and reference beams is implemented and calibrated to ensure accurate polarization reconstruction. Retardation and principal-stress distributions are then computed using the stress–optic law. The method is validated on a 500-µm-thick glass sample containing nine regions processed at different laser powers. Matched regions of interest are analyzed using SPDHM and a conventional digital polariscope for reference, and the resulting mean stress values are directly compared. SPDHM captures the laser-power-dependent stress variation across all regions. Numerical refocusing further enables depth-selective inspection of top- and bottom-surface features without mechanical adjustment, supporting high-speed in-line stress monitoring and process optimization.

1. Introduction

Glass-based interposers have emerged as a promising platform for advanced semiconductor packaging, offering high electrical performance for high-bandwidth operation and reliable metal interconnect integration [1]. These interposers are commonly fabricated using a laser-induced wet etching process. The laser modification process, however, can introduce residual stress that degrades mechanical reliability and yield through cracking. Therefore, the non-destructive stress metrology is important for process optimization and in-line inspection [2]. In this work, single-exposure polarization digital holographic microscopy (SPDHM) was developed for quantitative stress characterization in laser-modified glass substrate [3]. The proposed system enables single-acquisition capture of polarization-resolved holograms, from which the complex optical field and its polarization information are retrieved, providing amplitude, phase, and stress-induced birefringence [4].

The method uses interference between reference and object waves under controlled polarization states to retrieve polarization-resolved complex fields and reconstruct the Jones vector components [4]; numerical propagation with the ASM reconstructs the complex field at the plane of interest, while the stress optic law is applied to obtain spatial maps of retardation and principal stress in transparent glass substrates [5].

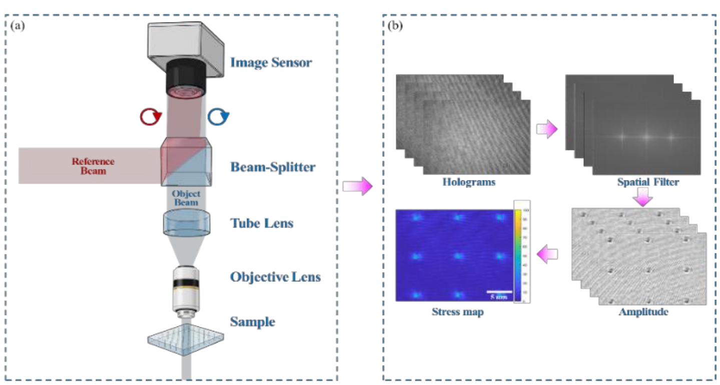

Figure 1. SPDHM stress metrology platform: (a) system schematic, (b) digital hologram processing and reconstruction results.

2. Results and Discussion

Figure 2(a) and (b) show the full-field stress maps of the laser-modified glass substrate. Both digital photoelasticity and polarization digital holography show nearly identical stress localization patterns concentrated around the laser-modified zone. The average stress across nine distinct regions fabricated with varying laser power is compared in Figure 2(c). Both methods exhibit a consistent positive correlation, demonstrating that increased laser power predictably elevates the induced residual stress. This strong agreement across all test cases further validates the accuracy of the proposed holographic approach. Figure 2(d) and (e) show scratches located on the top and bottom surfaces of the glass, respectively, demonstrating multi-plane visualization through numerical refocusing.

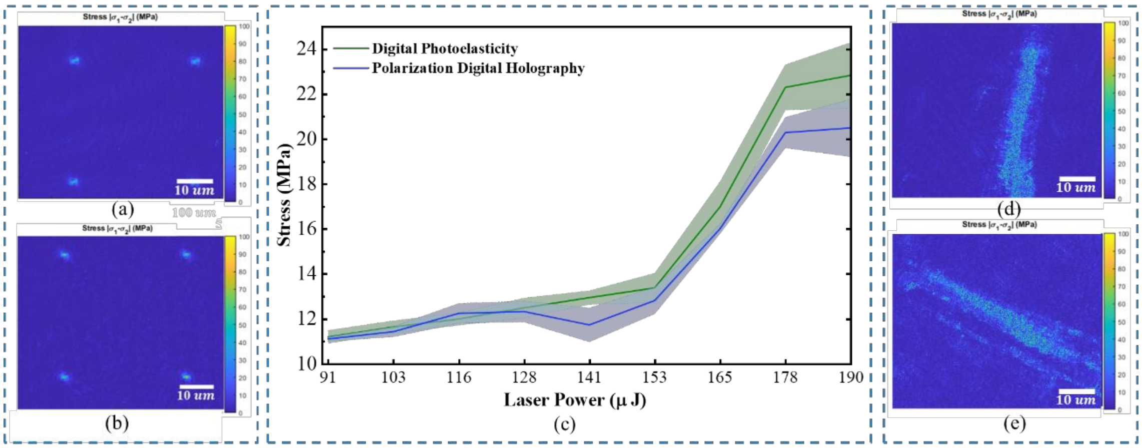

Figure 2. Residual stress measurement in laser-modified glass substrate using (a) conventional digital polariscope system, (b) proposed SPDHM. (c) Comparison of quantitative stress variation with varying laser power for both methods. (d-e) Numerically refocused view of scratches on the top and bottom surfaces of the glass substrate.

The difference between SPDHM and digital photoelasticity is summarized using the mean absolute percentage error (MAPE). Nine matching regions of interest (ROIs) were selected, and the mean stress within each ROI was extracted from the digital polariscope map as the reference value and from the holography map as the experimental value. For each region, an absolute percentage error was computed as the absolute difference between the two ROI means normalized by the reference mean. The MAPE was then obtained by averaging the nine absolute percentage errors. The reported MAPE of 4.95%, therefore, represents the average relative deviation between the proposed method and the reference across the tested laser-power conditions. To verify the refocusing capability of the system, scratches were introduced on the top and bottom glass surfaces. A single acquisition enabled each plane to be brought into focus through numerical propagation, demonstrating a key advantage of the holography-based approach by avoiding mechanical adjustment.

3. Conclusions

A single-exposure polarization digital holographic microscope (SPDHM) was successfully developed for non-destructive, full-field residual-stress metrology in laser-modified glass substrates. The proposed approach enables single-shot reconstruction of the complex optical wavefront and polarization state, from which quantitative retardation and principal-stress maps are derived via stress-induced birefringence analysis. Validation against a conventional digital polariscope on nine regions processed at different laser powers shows strong agreement, with a mean absolute percentage error of 4.95% relative to the reference measurements. Furthermore, the holography-based framework enables numerical refocusing, allowing multi-plane inspection from the same acquisition. This capability facilitates correlation of stress distributions with surface and subsurface features without mechanical refocusing, supporting efficient in-line monitoring and process optimization for transparent substrate manufacturing.

References

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